

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
// You are using Java
import java.util.Scanner;
```

```
class DotException extends Exception {
    public DotException(String message) {
        super(message);
    }
}
```

```
class AtTheRateException extends Exception {
    public AtTheRateException(String message) {
        super(message);
    }
}
```

```
class DomainException extends Exception {
    public DomainException(String message) {
        super(message);
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String email = scanner.nextLine();
        scanner.close();

        try {
            validateEmail(email);
            System.out.println("Valid email address");
        } catch (DotException e) {
            System.out.println("DotException: " + e.getMessage());
        }
    }
}
```

```

        System.out.println("Invalid email address");
    } catch (AtTheRateException e) {
        System.out.println("AtTheRateException: " + e.getMessage());
        System.out.println("Invalid email address");
    } catch (DomainException e) {
        System.out.println("DomainException: " + e.getMessage());
        System.out.println("Invalid email address");
    }
}

```

```

static void validateEmail(String email) throws DotException,
    AtTheRateException, DomainException {
    if (email.startsWith(".") || email.endsWith(".") || email.startsWith("@") ||
        email.endsWith("@")) {
        throw new DotException("Invalid Dot usage");
    }
}

```

```

int atCount = 0;
for (char ch : email.toCharArray()) {
    if (ch == '@') atCount++;
}
if (atCount != 1) {
    throw new AtTheRateException("Invalid @ usage");
}

```

```

for (int i = 1; i < email.length(); i++) {
    if ((email.charAt(i) == '.' && email.charAt(i - 1) == '.') ||
        (email.charAt(i) == '@' && email.charAt(i - 1) == '@')) {
        if (email.charAt(i) == '.') {
            throw new DotException("Invalid Dot usage");
        } else {
            throw new AtTheRateException("Invalid @ usage");
        }
    }
}
}

```

```

int atIndex = email.indexOf('@');

```

```

String afterAt = email.substring(atIndex + 1);
int dotAfterAtCount = 0;
for (char ch : afterAt.toCharArray()) {
    if (ch == '.') dotAfterAtCount++;
}

```

```
}  
if (dotAfterAtCount != 1) {  
    throw new DotException("Invalid Dot usage");  
}  
  
int lastDotIndex = email.lastIndexOf('.');  
if (lastDotIndex < atIndex) {  
    throw new DotException("Invalid Dot usage");  
}  
String domain = email.substring(lastDotIndex + 1);  
if (!(domain.equals("in") || domain.equals("com") || domain.equals("net") ||  
domain.equals("biz"))) {  
    throw new DomainException("Invalid Domain");  
}  
}  
}
```

Status : Correct

Marks : 10/10